GSE 518
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Suggested Solutions to Problem Set 2

1.
(a) The sample space is given by $S = \{1, 2, 3, 4, 5, 6\}$. $A = \{5, 6\}$. Therefore, $P(A) = 1/3$.
(b) The area of the square is 4. The area of the square for which the sum of the coordinates is less than 1/2 is given by $4 - \frac{3}{2}\frac{3}{2}\frac{1}{2} = \frac{23}{8}$ (See attached graph). Therefore, $P(A) = \frac{23}{8}\frac{1}{4} = 23/32$.

2.
(a) $S = \{(H), (TH), (TTH), (TTTH), \ldots\}$.
(b) Each toss is independent of each other and therefore $P((TTH)) = P(T) P(T) P(H) = \frac{1}{2}\frac{1}{2}\frac{1}{2} = 1/8$
(c) $P(x \text{ tosses are needed to get first head}) = P(T^{x-1}H) = (1/2^{x-1})(1/2) = 1/2^x$.

3.
(a) In this case the sample space is given by $S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$. From here we can see that $P(A) = 1/2$, $P(B) = 1/2$, $P(C) = 1/2$, $P(A \cap B) = 1/4$, $P(A \cap C) = 1/4$ and $P(B \cap C) = 1/4$. Since $P(A \cap B) = P(A) P(B) = 1/4$, $P(A \cap C) = P(A) P(C) = 1/4$ and $P(B \cap C) = P(B) P(C) = 1/4$, these three events are pairwise independent.
(b) In order for the events A, B and C to be mutually independent we need $P(A \cap B \cap C) = P(A) P(B) P(C)$. We have that $P(A \cap B \cap C) = 0$ because if we get an odd face in both dice we cannot have an odd sum (i.e., $A \cap B \cap C = \phi$). However, $P(A) P(B) P(C) = 1/8$. Therefore, since $P(A \cap B \cap C) \neq P(A) P(B) P(C)$ the events are not mutually independent.
(c) We can calculate the probability of each event directly from the sample space. We have $P(D) = 1/6$, $P(E) = 1/2$ and $P(F) = 1/3$. Given that $A \cap B \cap C = \{(4,4)\}$, we have that $P(D \cap E \cap F) = 1/36$. Hence, $P(D) P(E) P(F) = (1/6)(1/2)(1/3) = 1/36 = P(D \cap E \cap F)$.
In order for the events D, E and F to be mutually independent we also need the events to be pairwise independent. Note that $P(E \cap F) = (\text{the sum is 7 or 8}) = 11/36$, while $P(E) P(F) = (1/2)(1/3) = 1/6$. Since $P(E \cap F) \neq P(E) P(F)$ the events E and F are not pairwise independent. Therefore, the events D, E and F are not mutually independent. (You could also show that $P(D \cap E) \neq P(D) P(E)$ or $P(D \cap F) \neq P(D) P(F)$).

4.
(a) The probability of selecting each box is $P(I) = 1/2$ and $P(II) = 1/2$. The conditional probabilities of choosing a red ball are $P(r|I) = 3/10$ and $P(r|II) = 6/10$. Finally, the probability that the ball drawn is red is given by

$$P(r) = P(r|I)\,P(I) + P(r|II)\,P(II) = \left(\frac{3}{10}\right)\left(\frac{1}{2}\right) + \left(\frac{6}{10}\right)\left(\frac{1}{2}\right) = \frac{9}{20}$$

(b) Using Bayes' rule

$$P(II|r) = \frac{P(r|II)\,P(II)}{P(r)} = \frac{(6/10)\,(1/2)}{(9/20)} = \frac{2}{3}$$

5. The probability of visiting each of the cities is given by $P(S) = 1/3$, $P(LV) = 1/3$ and $P(NY) = 1/3$. The probability of rain in each state is given by $P\left(R^S\right) = 1/3$, $P\left(R^{LV}\right) = 1/10$ and $P\left(R^{NY}\right) = 1/5$. It is fair to assume that the event of raining in any city is independent of whether your friend visits the city or not. Hence we have $P\left(S \cap R^S\right) = P(S)\,P\left(R^S\right) = 1/9$, $P\left(LV \cap R^{LV}\right) = P(LV)\,P\left(R^{LV}\right) = 1/30$ and $P\left(NY \cap R^{NY}\right) = P(NY)\,P\left(R^{NY}\right) = 1/15$. Let W denote the event that your friend got wet (i.e., that she went to a city in which it rained). Given that the events S, LV and NY are disjoint, the events $S \cap R^S$, $LV \cap R^{LV}$ and $NY \cap R^{NY}$ are also disjoint. Therefore,

$$\begin{aligned} P(W) &= P\left(S \cap R^S\right) + P\left(LV \cap R^{LV}\right) + P\left(NY \cap R^{NY}\right) \\ &= \frac{1}{9} + \frac{1}{30} + \frac{1}{15} = \frac{19}{90} \end{aligned}$$

Using the definition of conditional probability we find

$$P(S|W) = \frac{P(S \cap W)}{P(W)} = \frac{P\left(S \cap R^S\right)}{P(W)} = \frac{(1/9)}{(19/90)} = \frac{10}{19}$$

where the second equality comes from the fact that your friend gets wet in Seattle only if it rained there. Similarly, we have

$$P(LV|W) = \frac{P\left(LV \cap R^{LV}\right)}{P(W)} = \frac{3}{19} \quad \text{and} \quad P(NY|W) = \frac{P\left(NY \cap R^{NY}\right)}{P(W)} = \frac{6}{19}$$

6. For some problems (like this one) is easier to solve for the *complement* of the event. Here we have $P(\textit{at least}$ one double six in n throws$) = 1 - P($no double six in n throws$)$. Since (or assuming) all tosses are independent, we have $P($no double six in n throws$) = P(\text{no double six})^n = (35/36)^n$. By solving $1-(35/36)^n > 0.5$ we find that we need $n > 24.6$, or at least 25 rolls.

FIGURE FOR QUESTION 1.B.

